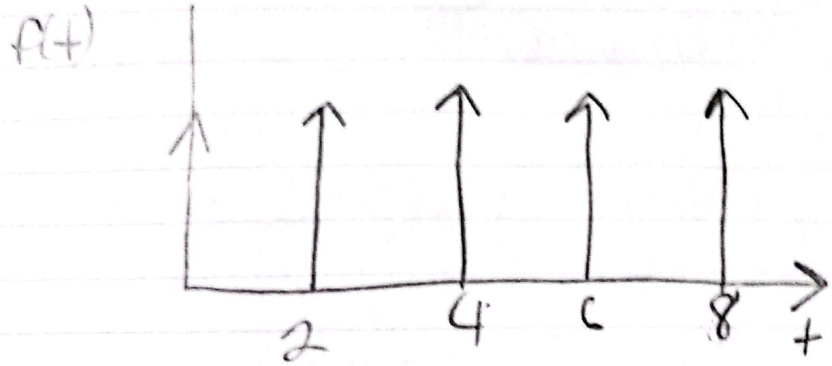
1)



$$= \int_0^T \frac{f(t)\, e^{-st}\, dt}{1 - e^{-st}} \qquad T = 2$$

$$= \int_0^2 f(t)\, e^{-st}\, dt$$

$$= 1$$

$$L\{f(t)\} = \frac{1}{1 - e^{-2s}}$$

$$\frac{d^2y(t)}{dt^2} + 6\frac{dy(t)}{dt} + 5y(t) = 2u(t)$$

$$y(0_-) = 1 \qquad y'(0_-) = -2$$

$$= s^2 Y(s) - sy(0_-) - y'(0_-) + 6sY(s) - 6y(0_-)$$

$$+ 5Y(s) = \frac{2}{s}$$

$$= Y(s)[s^2 + 6s + 5]$$

$$= \frac{2}{s} - 12 + s - 2$$

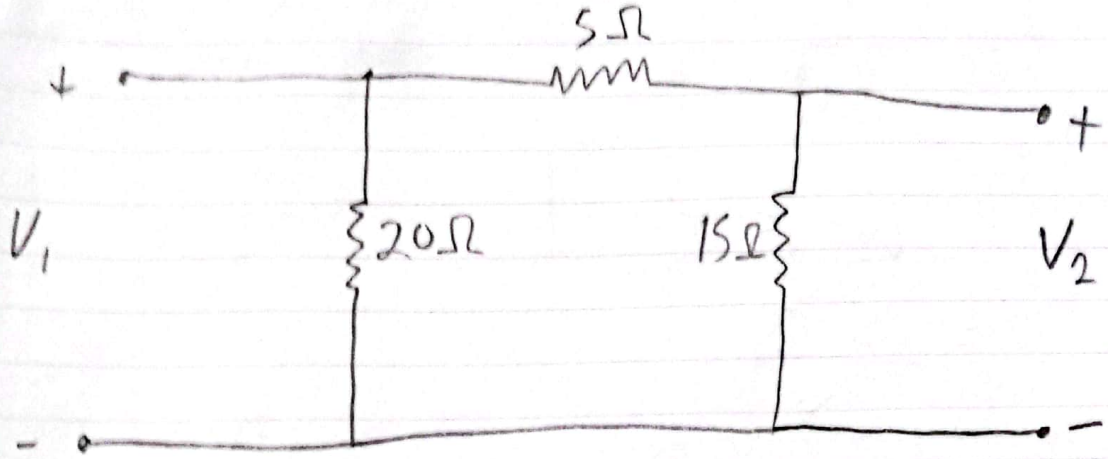
$$= \frac{s^2 - 14s + 2}{s}$$

$$Y(s) = \frac{s^2 - 14s + 2}{s[s^2 + 6s + 5]} = \frac{s^2 + 14s + 2}{s(s+1)(s+5)}$$

$$Y(s) = \frac{.4}{s} - \frac{4.25}{s+1} + \frac{4.85}{s+5}$$

$$\boxed{y(t) = (.4 - 4.25e^{-t} + 4.85e^{-5t})u(t)}$$

3)



$$V_1 = Z_{11} I_1 + Z_{12} I_2$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2$$

$$Z_{11} = \frac{V_1}{I_1} = \frac{I \cdot 20}{40} = \frac{I_1}{2}$$

$$V_1 = \frac{20 I_1}{2}$$

$$\boxed{Z_{11} = 10\,\Omega}$$

$$Z_{12} = \frac{V_1}{I_2} \qquad I_2' = \frac{15}{40} I_2$$

$$= V_1 I_2' \qquad = \frac{3}{8} I_2$$

$$= 20 \cdot \frac{3}{8} I_2$$

$$\boxed{Z_{12} = 7.5\,\Omega}$$

$$Z_{21} = \frac{V_2}{I_1}$$

$$I_1'' = \frac{I_1 \cdot 20}{40}$$

$$= \frac{I_1}{2}$$

$$V_2 = 15\, I_1''$$

$$= \frac{15 \cdot I_1}{2}$$

$$\boxed{Z_{21} = 7.5\ \Omega}$$

$$Z_{22} = \frac{V_2}{I_2}$$

$$I_2'' = \frac{I_2 \cdot 25}{40}$$

$$= \frac{5}{8} I_2$$

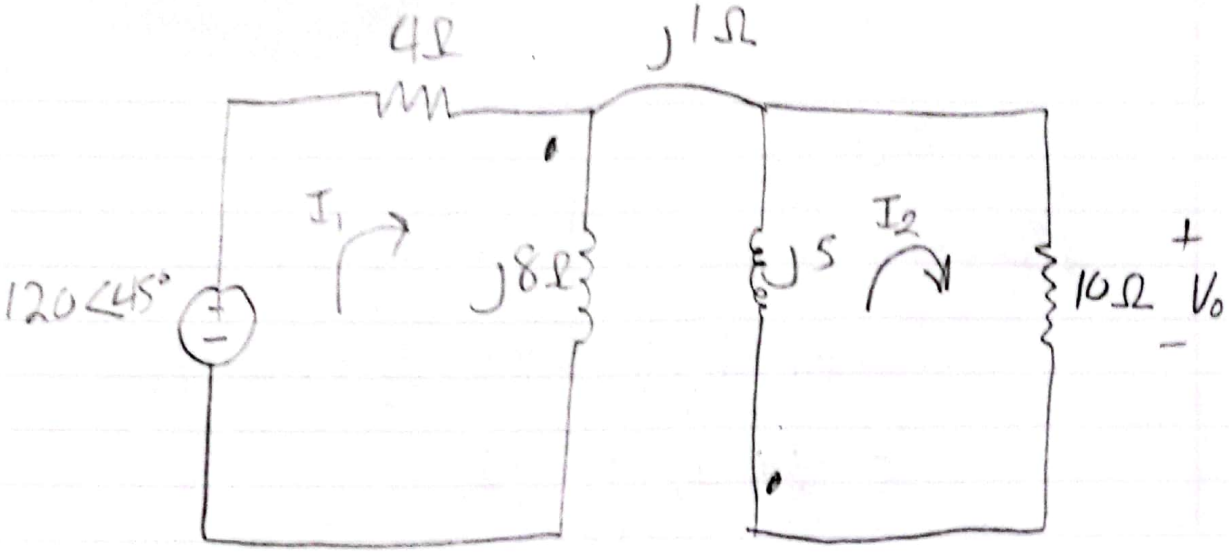
$$V_2 = I_2'' \cdot 15$$

$$= \frac{75}{8} i_2$$

$$Z_{22} = \frac{75}{8}\ \Omega$$

$$\boxed{Z_{22} = 9.375\ \Omega}$$

CamScanner



KVL loop 1

$$-120\angle 45 + 4 + 8jI_1 + 1jI_2 = 0$$

$$120\angle 45^\circ = (4+8j)I_1 + 1jI_2$$

KVL loop 2

$$10I_2 + 5jI_2 + 1jI_1 = 0$$

$$(10+5j)I_2 + 1jI_1 = 0$$

$$I_1 = \frac{(-10+5j)I_2}{j}$$

$$I_1 = (-5+10j)I_2$$

So

$$120\angle 45 = (4+8j)I_1 + 1j\,I_0$$

$$= (4+8j)(-5+10j)I_2 + 1j \cdot I_2$$

$$= -100 I_2 + j I_2$$

$$120\angle 45 = (-100+j) I_2$$

$$I_2 = \frac{120\angle 45}{-100+j}$$

$$= 1.2\angle -134.427 \text{ A}$$

$$V_0 = 10 \cdot I_2$$

$$= 10 \cdot 12\angle -134.427$$

$$V_0 = 12\angle -134.427 \text{ V}$$